2017-Ph-H1-Q1

Section: Our Dynamic Universe
Topic: Motion, Equations and Graphs

A graph of velocity against time for an object is shown.
Which row in the table describes the acceleration and the displacement of the object during the time interval shown?

We're given a **velocity–time graph**.

From such a graph:

- **Acceleration** is the **slope** of the graph
- **Displacement** is the **area under** the graph

Let's work through it step by step:

🔍 Acceleration:

If the graph is a **straight line sloping upward**, the velocity is increasing at a constant rate → **positive, constant acceleration**.

If the line is **flat**, acceleration is **zero**.

If it's **curving**, acceleration is **changing**.

From the source, the graph **has a straight, upward slope** from time 0 to 4 s — so:

✅ **Acceleration is constant.**

🔍 Displacement:

The **area under a v–t graph** gives displacement.
If the velocity is increasing, the area increases — meaning:

✅ **Displacement is increasing.**

More specifically, since velocity is always **positive** and increasing, the object keeps moving **forward** and covering **more distance**.

✅ Final Answer: A — Acceleration: constant, Displacement: increasing

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🧠 Revision Tips:

- Velocity–time graph:
  - **Slope = acceleration**
  - **Area under graph = displacement**
- Straight diagonal line → constant acceleration
- Curved line → changing acceleration
- Flat line → zero acceleration
